CSCI.4923 Capstone in Interprofessional Informatics

Returns Comparison of Constant Mix and Buy-Hold Strategies

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Contents

List of Figures

List of Tables

1 Abstract

This study aims to compare the performance of the buy-hold strategy and the 40% bond/60% stock rebalancing strategy using historical simulation from August 2002 to August 2022, a 20 year period. I will also compare the returns of different asset allocations, which includes small cap (IWM), technology (QQQ), large cap (SPY), and bonds (TLT). For each strategy, contributions will be made monthly. For the buy-hold strategy, stocks accumulated each month will never be sold until the end of August 2022. For the 40/60 constant mix strategy, rebalancing will take place at the end of each month until end of August 2022.

2 Introduction

2.1 Background

Financial literacy is important. However, it is difficult to grasp. Lusardi, A. & Mitchell, O. (2011) reported in 2011 that financial literacy is particularly low among the young, women, and the less-educated. Moreover, their study suggests that Hispanics and African-Americans score the least well on financial literacy concepts.

This includes retirement planning. In 2021, the Federal Reserve Board reports that three-fourths of non-retired adults had at least some retirement savings. Among those with retirement savings, these savings were most frequently in defined contribution plans, such as a 401(k) or 403(b), with 55% of non-retired adults having money in such a plan. In the Harris Poll's 2021 Financial Literacy and Preparedness Survey, 12% of adults said they worried about retiring without having set aside enough money.

Understanding retirement/long-term investing and how to invest is beneficial in building wealth and financial freedom over time. Lusardi, A. & Mitchell, O. (2011) found that people who score higher on the financial literacy questions are much more likely to plan for retirement. In addition, Clark, R., Lusardi, A., & Mitchell, O. (2017) found that the most financially knowledgeable investors held 18% points more stock than their least knowledgeable counterparts, could anticipate earning 0.08% per month more in excess returns, had 40% higher portfolio volatility, and held portfolios with about 38% less risk, as compared with less financially knowledgeable investors.

Nowadays, 401k contribution plans allow individuals to actively participate in customizing and electing their own investment strategies and asset allocations.

It can be quite challenging to decide which strategies and assets to elect, especially since the stock market is characteristically volatile and dependent on many factors. In addition, there are many investment strategies and assets choices, all with various risks and returns. One type of investment strategy is rebalancing strategy. Rebalancing strategy reduces risk and enhances returns through periodically buying and selling assets to maintain the desired allocation. Common simple rebalancing strategies include buy-hold and constant-mix.

2.2 Buy-Hold strategy

Traditionally, investors would try to maximize their gains and minimize their losses through buying stock when the stock price is low and selling stocks when the stock price is high. However, this “buy low and sell high” strategy requires intuition, knowledge, and timing to execute successfully. A simpler, but also successful strategy is the Buy-hold strategy. The Buy-hold strategy is a passive, long-term investment strategy where one buys stocks and holds on to the stock for a long period of time. Buy-hold strategy allows the individual to invest successfully with little management. The Buy-hold strategy’s success lies in the long-term growth of stocks over the years despite market fluctuations. So far, the market has been able to recover from declines and overall has positive annual returns. One disadvantage of this strategy is that it may take time to see growth. There is a large opportunity cost to invest in a stock long-term instead of investing in a stock short-term may give higher, quicker, returns. In this study, dollar cost averaging Buy-Hold strategy will be used. Dollar cost averaging is contributing a fixed amount to an investment on a regular schedule. In this case, a contribution of $1,000 dollars monthly will be made to the investment.

2.3 Constant-mix strategy

Constant-mix is an investment strategy that where one buys a ratio of different assets and maintains that ratio. A common constant-mix that will be used in this study is ratio is 60% stocks and 40% bonds. For instance, for the 60% stocks/40% bonds, if an individual wants to invest $1000, the individual is $400 bonds and $600 stocks. After a period, the individual will evaluate the value of the stocks and bonds. If the value of stocks is over 60% of the total value of the investments, then the individual will sell stocks back to 60%

and use the difference to buy bonds. If the value of stocks is under 60% of the total value of the investments, then the individual will sell bonds back to 40% and use the difference to sell stocks. This technique is known as rebalancing. Through rebalancing, constant mix allows one to practice the "buy low and sell high" strategy. The asset that performs the best is sold to buy the stock that performs worst. Constant-mix success lies in the volatility of the market, by taking advantage of market declines and selling during market highest. One disadvantage of constant-mix is that the strategy is more involved and requires constant adjustment to maximize gains. There are many ways to rebalance, but calendar rebalancing will be practiced in this study. Calendar rebalancing is a simple rebalancing approach that seeks rebalancing at fixed intervals such as quarterly or monthly. In this case, rebalancing will be done at the beginning of each month. In addition, contributions of $400 bonds and $600 stocks will be made at the beginning of each month as well.

2.4 Asset allocation

2.5 Literature Review

There are numerous studies conducted over the years about performance of constant mix and buy-hold strategies. Tower, E. & Gokcekus, O. (2002) evaluates a long-term buy-hold strategy for the SP index. In the paper, they calculated the real rate of return from purchasing the SP 500 index from 1871 through 2001. They assumed that the investor invests yearly, consuming dividends, and never selling until the year 2001. The results showed the expected return was around 3% yearly. Bertrand, P. & Prigent, J-L. (2022) compares the performance of constant mix strategy and buy-hold strategy. Their study found that the constant mix payout is more often higher than the buy-and-hold payout, but not significantly. There are times when buy-hold outperforms better than constant mix in the right conditions. Hilliard, JE & Hilliard, J. (2015) compares the returns from constant mix and buy and hold portfolios with the same asset. They found that constant mix reduces volatility. On the other hand, the buy and hold strategy had higher returns of the asset. O'Brien, TJ. (2020) investigates which investors will be better off with buy-hold or constant mix. They recommend that risk-tolerant investors with levered equity positions are better off with buy-hold than constant mix. Conversely, traditional investors, with positive allocations to both equity and fixed income, are better off with constant mix than with buy-hold. Qian, EE. (2014) compares the expected value and returns from constant mix and buy hold portfolios with the same stocks using terminal wealth as a measure. They compared expected value and variance for constant mix and buy-and-hold strategies using terminal wealth as a measure. They found that constant mix portfolios are more likely to have better risk-adjusted terminal wealth than buy-and-hold portfolios. Boscaljon, B. & Sun, L. (2006) compares constant mix, constant proportion portfolio insurance (CPPI) strategies and the buy-and-hold strategy for 5-, 10-, 20-, and 30- year periods. They also recommend constant mix and buy-hold portfolios for individuals willing to take on risk to get higher returns.

From these studies, it can be concluded that the constant mix strategy is more resistant to risk then the buy-hold strategy. However, the returns and the performance of each strategy may vary due to many factors. In this study, this conclusion will be tested, but using different asset allocations and done over a 20 year period.

3 Methodology

3.1 Return

Number of stocks is calculated using contributions (money invested) and the price the stock was bought.

$$stocks = \frac{contributions}{stock\_price}$$

The returns is calculated using number of stocks multiplied by the final stock price.

$$returns = stocks * final\_price$$

3.2 Rate of Return

Rate of Return (%) or Return on Investment (ROI) is calculated using total contributions and value of portfolio.

$$ROI = \frac{(portfolio\_value - total\_contributions)}{total\_contributions}$$

3.3 Max Drawdown

Max Drawdown (%) identifies the maximum decline of return on investment.It indicates the largest % loss from the portfolio. It is calculated using the highest return and the lowest return before the investment rises to a peak. This a common measure to measure how strategies deal with volatility and the largest expected % loss from portfolio or asset.

$$MDD(T) = \frac{P - L}{P}$$

P = peak value

L = lowest value before new peak

3.4 Algorithm: Buy-Hold strategy

Input:

price: stock price (of that month)

c: contributions (in this case, c = $ 1,000)

Output:

$total\_shares$

Method:

For each month,

1. calculate amount of shares, S.
2. add S to $total\_shares$.

Calculate the final return based on amount of $total\_shares$ **accumulated over the months.**

3.5 Algorithm: Constant-Mix

Input:

s_price: stock price (of that month)

b_price: bond price (of that month)

c: contributions (in this case, c = $ 1,000)

Output:

$total\_s\_shares$

$total\_b\_shares$

Method:

For each month,

1. allocate 40% of the c to bonds and 60% of the c to stock. (in this case, that is \$400 to bonds and \$600 to stocks.)
2. buy bonds shares, B and stock shares, S based on the allocation. Add B to $total\_b\_shares$ and S to $total\_s\_shares$.
3. calculate total portfolio value.
4. calculate the % of value in stocks and % of value in bonds inside the portfolio.
   (a) **if % value in stocks is greater than 60%**, then sell stock shares, s until % value is back to 60%. Use the money from selling s to buy bond shares b. $total\_s\_shares$ is deducted s amount and b is added to $total\_b\_shares$.
   (b) **if % value in stocks is less than 60%**, then sell bond shares, b until % value of bond shares is back to 40%. Use the money from selling the bond shares to buy stock shares s. $total\_b\_shares$ is deducted b amount and $total\_s\_shares$ increases s amount.
   (c) **if % value in stocks is equals 60%**, then no action is needed this month.

Calculate the final return based on amount of $total\_s\_shares$ **and** $total\_b\_shares$ **accumulated over the months.**

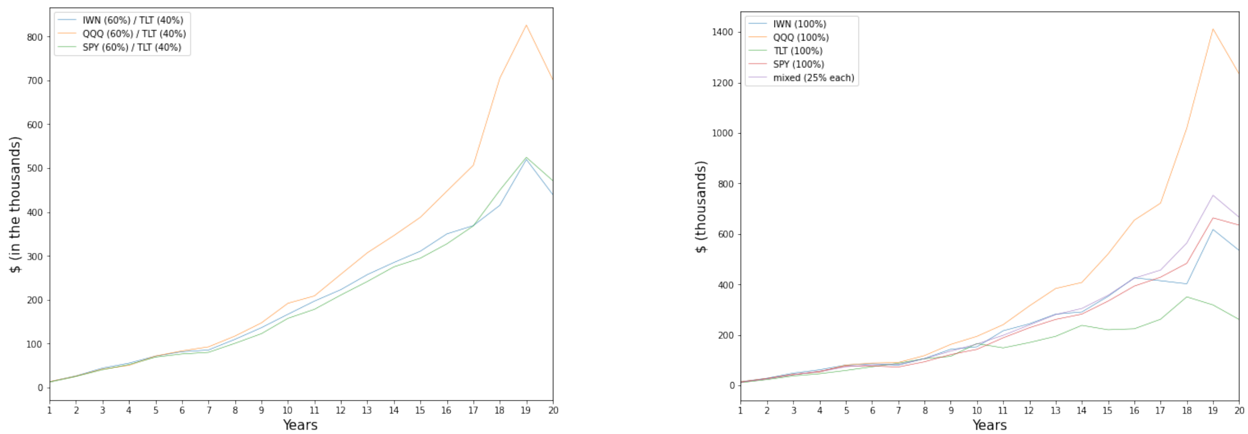
4 Dataset Description

5 Results

Table 1: Constant Mix vs. Buy-Hold strategies Final Return Summary

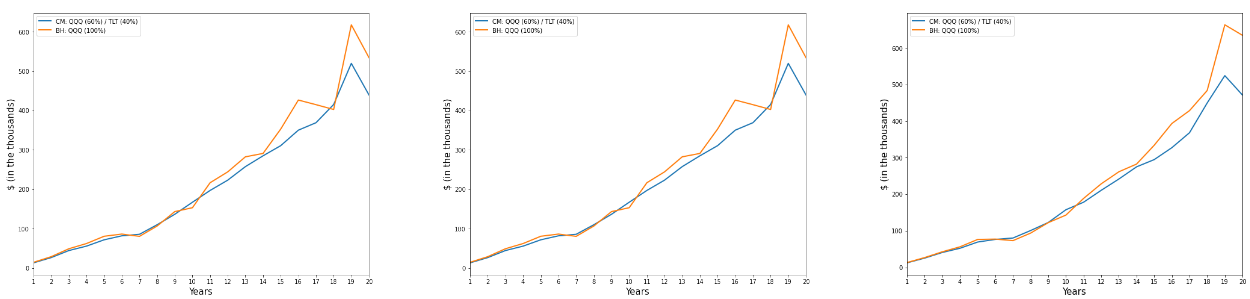
Strategies	Asset Allocation	Final Portfolio Value	Final Rate of Return	Max Drawdown (yearly)
Buy-Hold	IWN (100%)	$534,367.66	2.23%	-41.68%
	QQQ (100%)	$1,232,712.10	5.14%	-37.42%
	TLT (100%)	$260,990.26	1.09%	-22.32%
	SPY (100%)	$634,919.74	2.65%	-39.33%
	IWN/SPY/QQQ/TLT (25% each)	$665,747.44	2.77%	-25.32%
Constant-Mix (60/40)	IWN/TLT (60%/40%)	$439,590.48	1.83%	-22.84%
	QQQ/TLT (60%/40%)	$701,383.22	2.92%	-26.13%
	SPY/TLT (60%/40%)	$471,430.20	1.96%	-20.22%

Table 1 shows a summary of final return of all strategies and asset allocation combinations after 20 years. All portfolios ended with a positive final rate of return. The buy-hold strategy with 100% QQQ asset allocation has highest final portfolio value, which is 5% final rate of return. The strategy and asset allocation with the lowest final portfolio value is Buy-Hold with 100% TLT, which has 1.09% rate of return. The rest of the portfolios have final rate of returns ranging from 1.83% to 2.92%. The max drawdown measures the riskiness of each strategy and asset allocation combination. It indicates the max % loss the portfolio has had over the 20 years. The results show that the portfolio with the largest max drawdown is Buy-Hold strategy with 100% IWN allocation, which has a max drawdown of -41.68%. The second largest is the Buy-Hold strategy with 100% QQQ allocation. It has a max drawdown of -39.33%. The third largest is the Buy-Hold strategy with 100% QQQ allocation, which has a max drawdown of -37.42%. The lowest max drawdown is -20.22% from the constant-mix strategy with SPY 60% and TLT 40%. The max drawdown from the constant-mix strategy portfolios is significantly lower that their buy-hold portfolio counterparts. The max drawdown from constant-mix portfolios range from -20.22% to -26.13%, while max drawdown of buy-hold portfolios range from -22.32% to -41.68%. The result of Table 1 show that constant-mix strategy of 60% stock and 40% bond may be a less risky strategy than buy-hold with 100% asset allocation. However, constant-mix strategy may hold back the returns of high performing assets, because of the investment of 40% to TLT. The asset QQQ was the highest performing asset that had high final portfolio values and high final rate return in both Buy-Hold and constant-mix portfolios. TLT was the lowest performing asset. All other portfolios and assets had comparable final portfolio values and rate of returns.



(a) Constant-Mix (b) Buy-Hold

Figure 1: Cumulative Returns: Asset Allocation Comparison



(a) QQQ (b) IWN (c) SPY

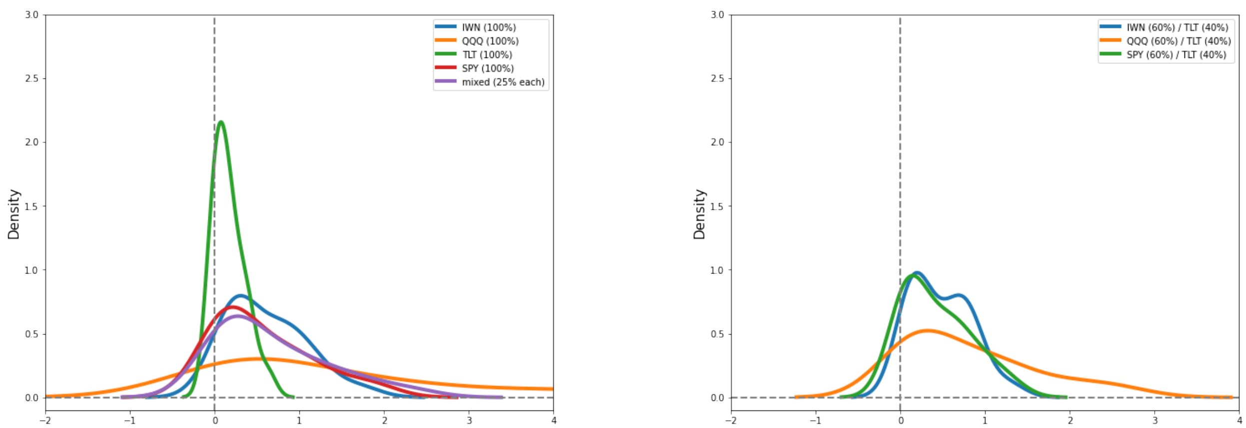
Figure 2: Cumulative Returns: Strategy Comparison

While max drawdown measures the largest loss, it does not account for the frequency of losses or the size of any gains. Therefore, a distribution of rate of returns over the years can help give a better understanding of expected rate of return yearly. Figure 1 and Figure 2 demonstrate the yearly cumulative returns for each portfolio over the 20 years. Figure 1 looks at the performance of assets of the strategy. In both Figure 1.a and 1.b shows that the returns of the all the assets is comparable within the first 10 years, but the asset QQQ starts outperforming the other assets significantly over time.

Figure 2 compares buy-hold and constant-mix strategy by analyzing the cumulative returns of the same asset. From these graphs, the rebalancing effect can be seen. When there is a great loss in returns or a dip in performance of buy-hold, constant-mix performs better and does not have the same level of loss or dip in performance. On the other hand, when there is great returns or increase in performance, constant-mix underperforms compared to their buy-hold counterparts. In general, the constant-mix portfolios shows less volatility in cumulative returns compared to their buy-hold counterparts. These graphs from Figure 2 demonstrates that the constant-mix portfolio more resistant to times of great loss but limits the returns during times of great gains.

Table 2: Distribution of Rate of Return - Yearly

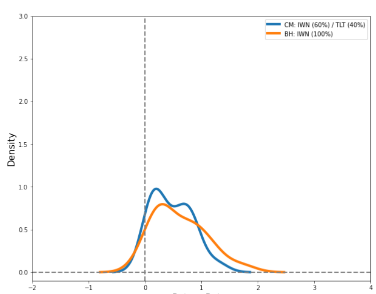
Strategies	**Asset Allocation**	**Mean**	**Median**	**Max**	**Min**	**Standard Deviation**	**Skewness**	**Kurtosis**
Buy Hold	IWN (100%)	0.61	0.50	1.71	-0.04	0.46	0.74	0.01
	QQQ (100%)	1.38	0.72	5.19	0.08	1.51	1.33	0.95
	TLT (100%)	0.17	0.12	0.62	-0.06	0.18	0.96	0.54
	SPY (100%)	0.56	0.35	1.91	-0.13	0.57	0.99	0.17
	IWN/SPY/QQQ/TLT (25% each)	0.68	0.44	2.30	-0.01	0.66	1.09	0.43
Constant Mix (60/40)	IWN/TLT (60%/40%)	0.48	0.44	1.28	0.02	0.35	0.54	-0.55
	QQQ/TLT (60%/40%)	0.81	0.59	2.62	0.05	0.78	1.02	0.15
	SPY/TLT (60%/40%)	0.42	0.33	1.30	-0.04	0.40	0.75	-0.43



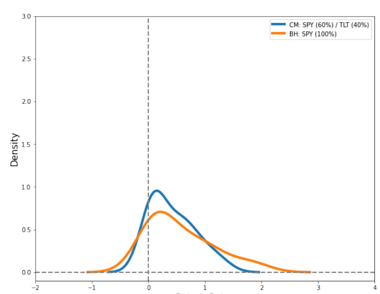
(a) Buy-Hold (b) Constant-Mix

Figure 3: Asset Allocation Density Plot Comparison

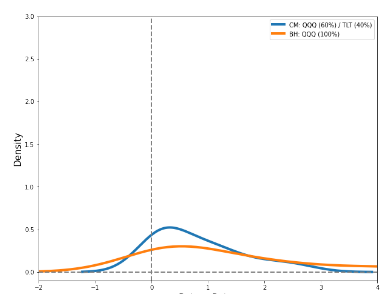
Table 2 summarizes the distribution of rate of return yearly. Figures 3a and 3b density plots display the distribution of rate of return yearly for each strategy and asset allocation studied. In Table 2, it can be observed that all strategies have low kurtosis, or thin-tailed with low outlier frequency. Constant-Mix IWN and SPY have negative kurtosis indicating that they have lighter tails than the normal distribution. Positive Skewness can also be observed in all of the strategies. It can be observed that the rate of return are more extreme and frequently positive. QQQ asset in buy-hold distribution has highest skewness, standard deviation, highest mean, highest median, highest maximum, and highest minimum of rate of return. On the other hand, QQQ asset in constant-mix displays a lower skewness, lower standard deviation, lower mean, lower median, and lower maximum, but a higher minimum rate of return. This effect can also be observed in both graphs from Figure 3. From Table 2, it can also be observed that the range of minimum rate of return yearly is from -0.13% to 0.08%, which does not vary much compared to the range of maximum rate of return. The range of maximum rate of return varies from 0.62% to 5.19%. Overall, the Table 2 and Figure 3 show less frequently extreme rates of return using constant-mix strategy. The constant-mix strategy will result lower extreme gains but also lower extreme losses. In addition, with assets that have higher standard deviation or more volatility such as QQQ, constant-mix can reduce the volatility of those assets.



(a) IWN



(b) SPY



(c) QQQ

Figure 4: Strategy Density Plot Comparison

Figure 4 compares each strategies distribution on the same asset. In all graphs from Figure 4, it can be observed that buy-hold has higher density of positive rates of return that are above 1% than the constant-mix strategy. While constant-mix has higher density of positive rates of return that are lower than 1%. As for negative rates of return, buy-hold has higher density of negative return rates then constant-mix in all graphs from Figure 4. Another noticeable feature is the bi-modality in some distributions, particularly in the constant-mix IWN distribution plot.

6 Conclusion

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